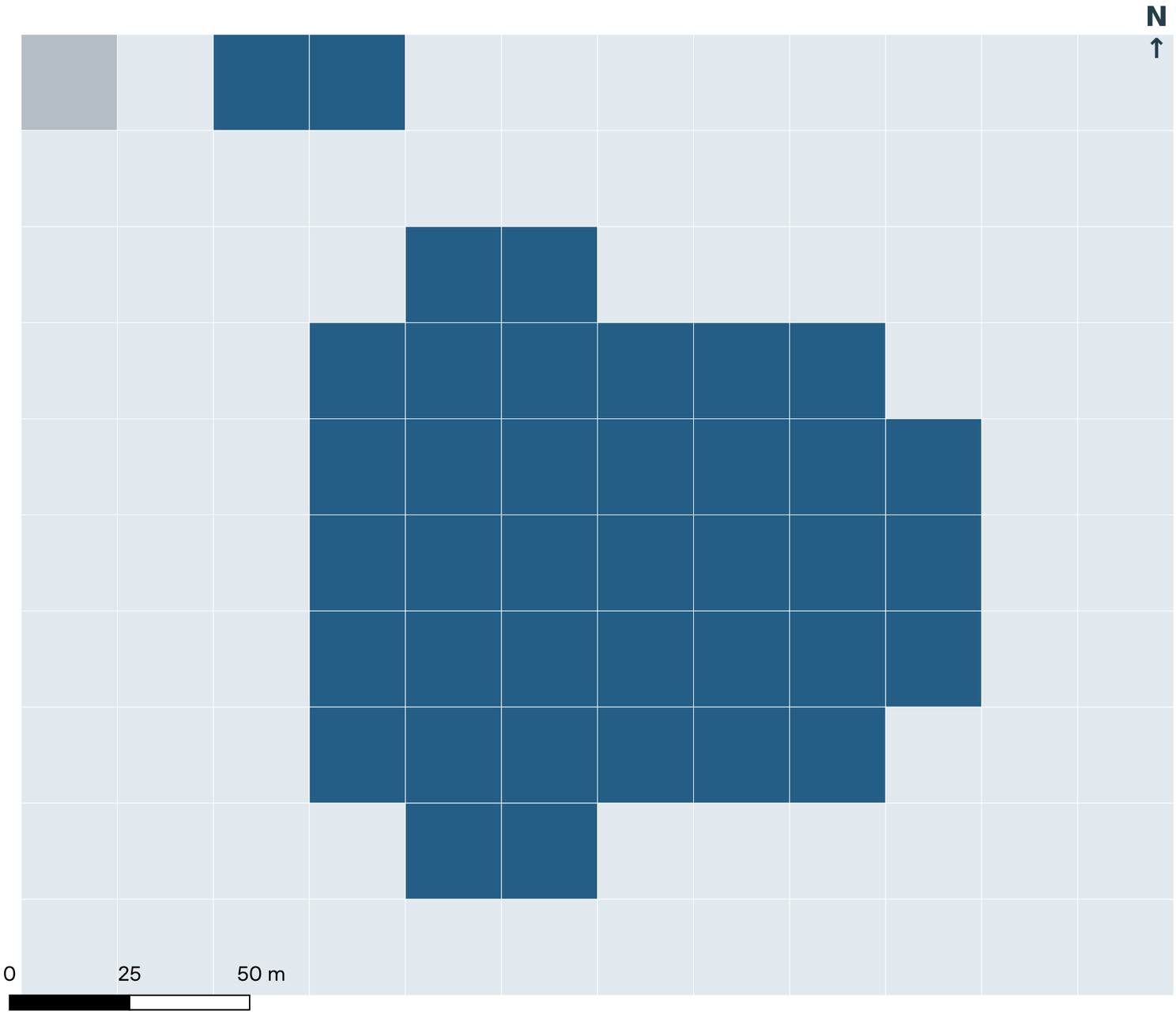
SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

01 · Flood intervention need

Synthetic test domain · not actual Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

- Intervention screen · depth ≥ threshold
- Depth below threshold
- Unknown / NoData

READ THIS MAP

Depth at or above the registered threshold identifies local need. Unknown cells remain unknown.

Below-threshold source-control areas are retained for investigation. Thesis §§5.1, 5.5.1; user threshold.

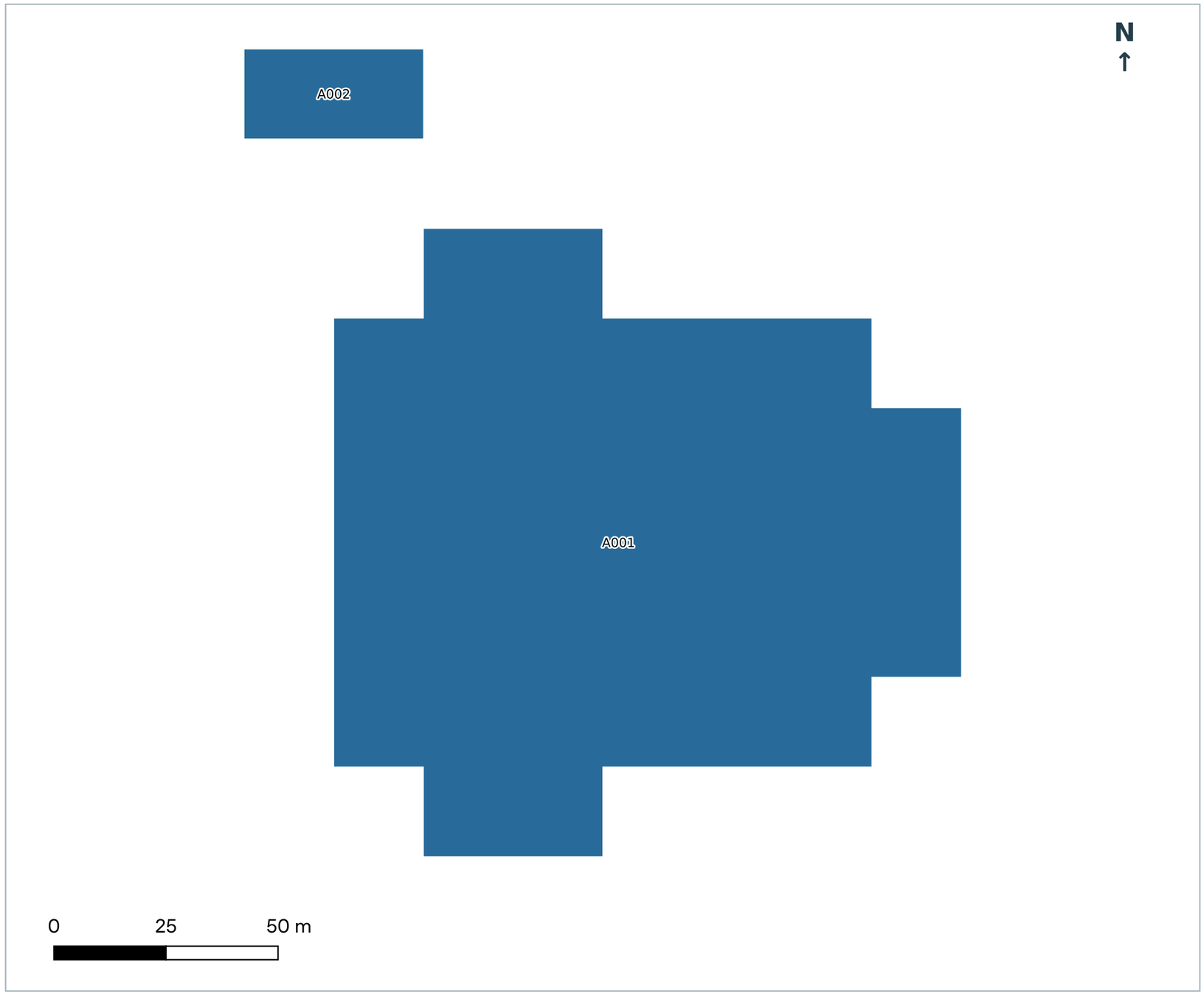
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: 20 × 20 m; source grid retained
Vertical datum: UNCONFIRMED
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_PLANNING_SRI5_V05
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

02 · Flood action areas

Synthetic test domain · not actual Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

Flood action area · investigate

READ THIS MAP

Connected flood-need areas for planning discussion. A001 etc. follow depth, then area order.

Read ownership shares, heat overlap and next action on click. These are not design modules. No flood benefit is estimated.

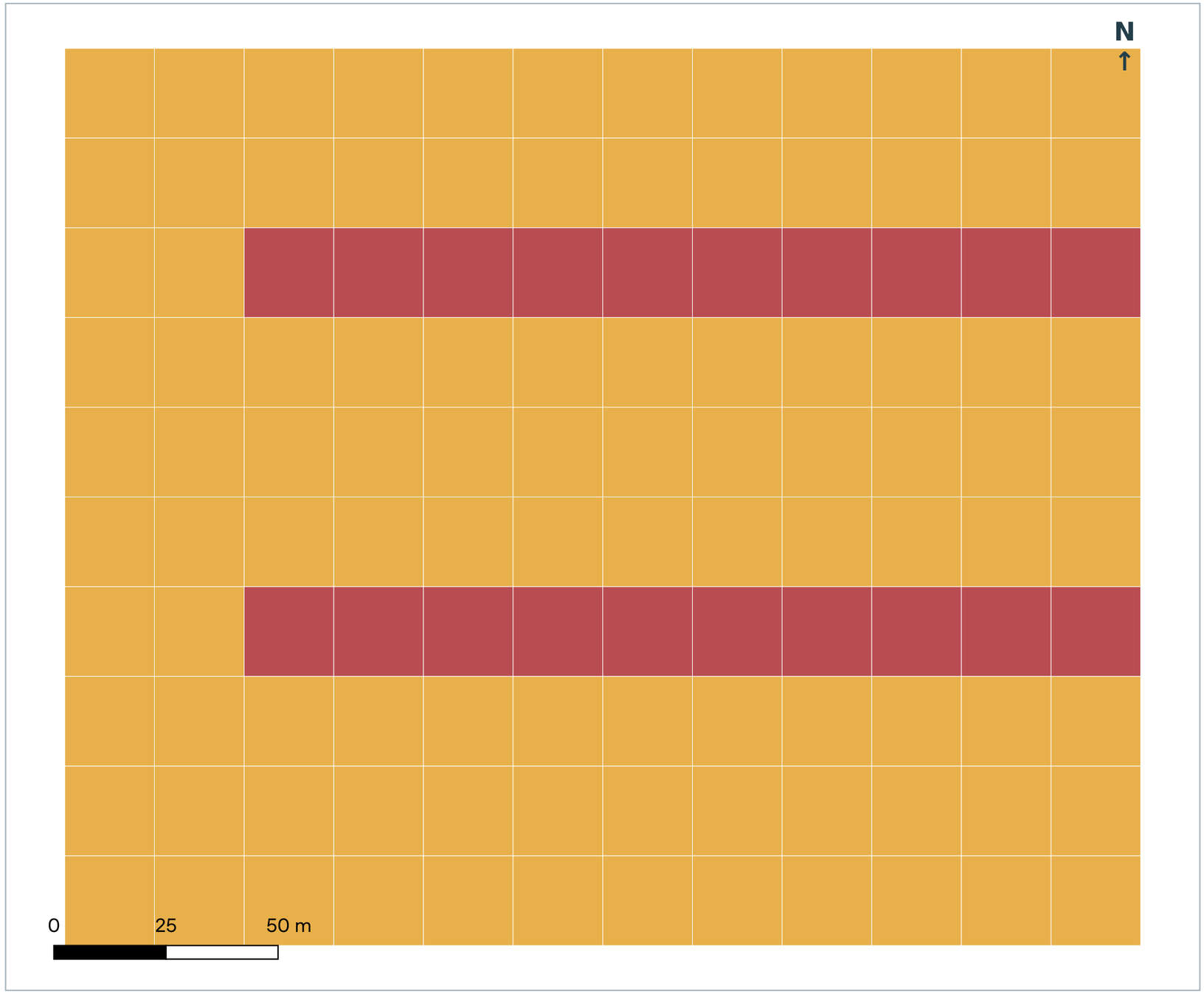
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: 20 × 20 m; source grid retained
Vertical datum: UNCONFIRMED
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_PLANNING_SRI5_V05
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

03 · Infiltration evidence and restrictions

Synthetic test domain · not actual Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

- Conditional · evidence / decision needed
- Fail · applicable gate not met

READ THIS MAP

Known prohibitions block infiltration.
Missing evidence stays conditional.

Lined detention and controlled outlets remain separate concepts.
Final clearance needs module design.
Thesis §§5.4.2, 5.5.2; Köln (2022).

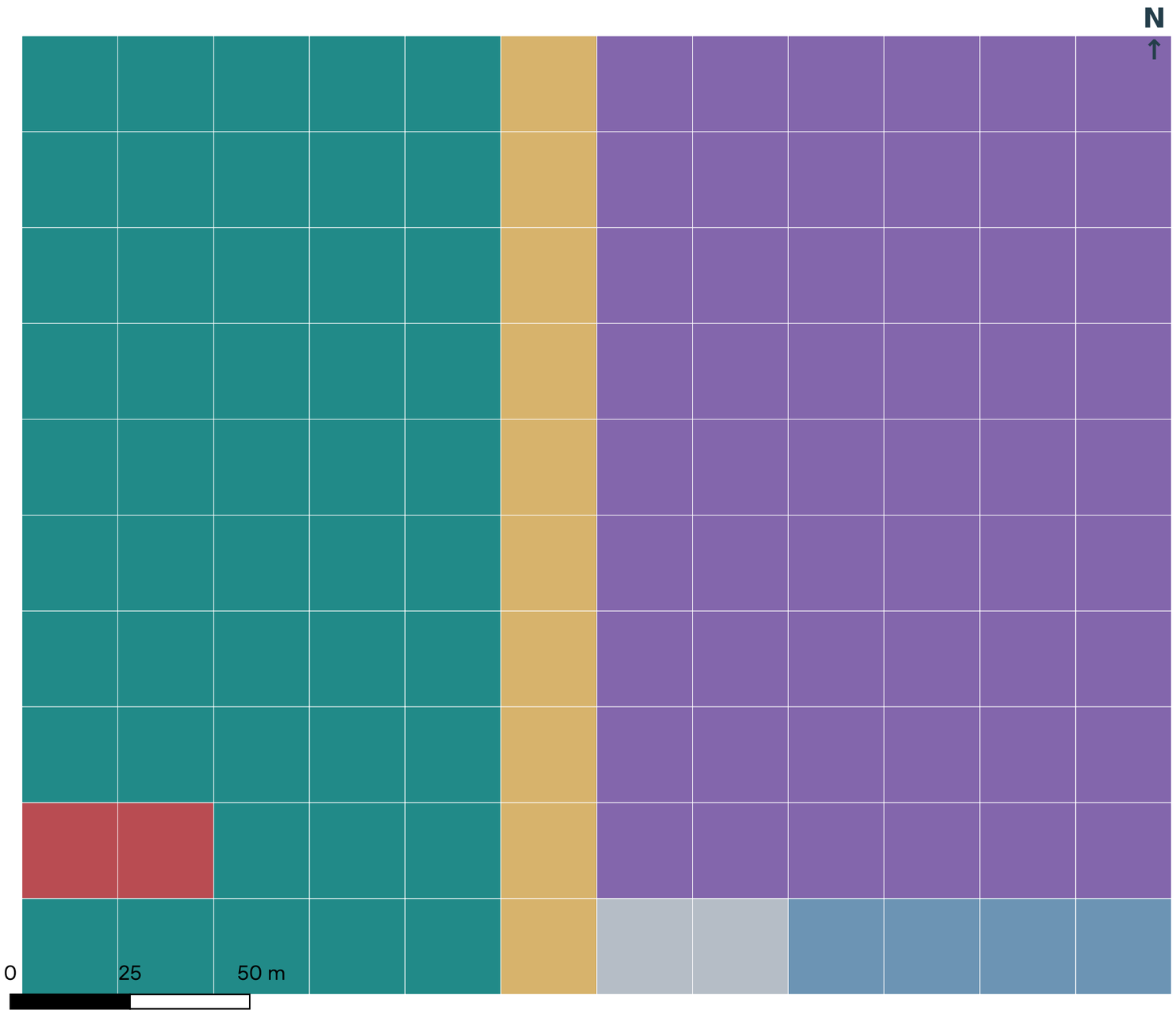
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: 20 × 20 m; source grid retained
Vertical datum: UNCONFIRMED
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_PLANNING_SRI5_V05
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

04 · Public and private delivery routes

Synthetic test domain · not actual Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

- Municipal ownership
- Private ownership
- Other public owner
- Mixed ownership · see area shares
- Conflicting overlapping ownership
- Unknown ownership

READ THIS MAP

Exact polygon area shares.
Private land is a partnership route.
Unknown ownership is retained.

Public use is not proof of ownership.
Click a unit for the next action.
Thesis §§5.4.1, 5.5.4.

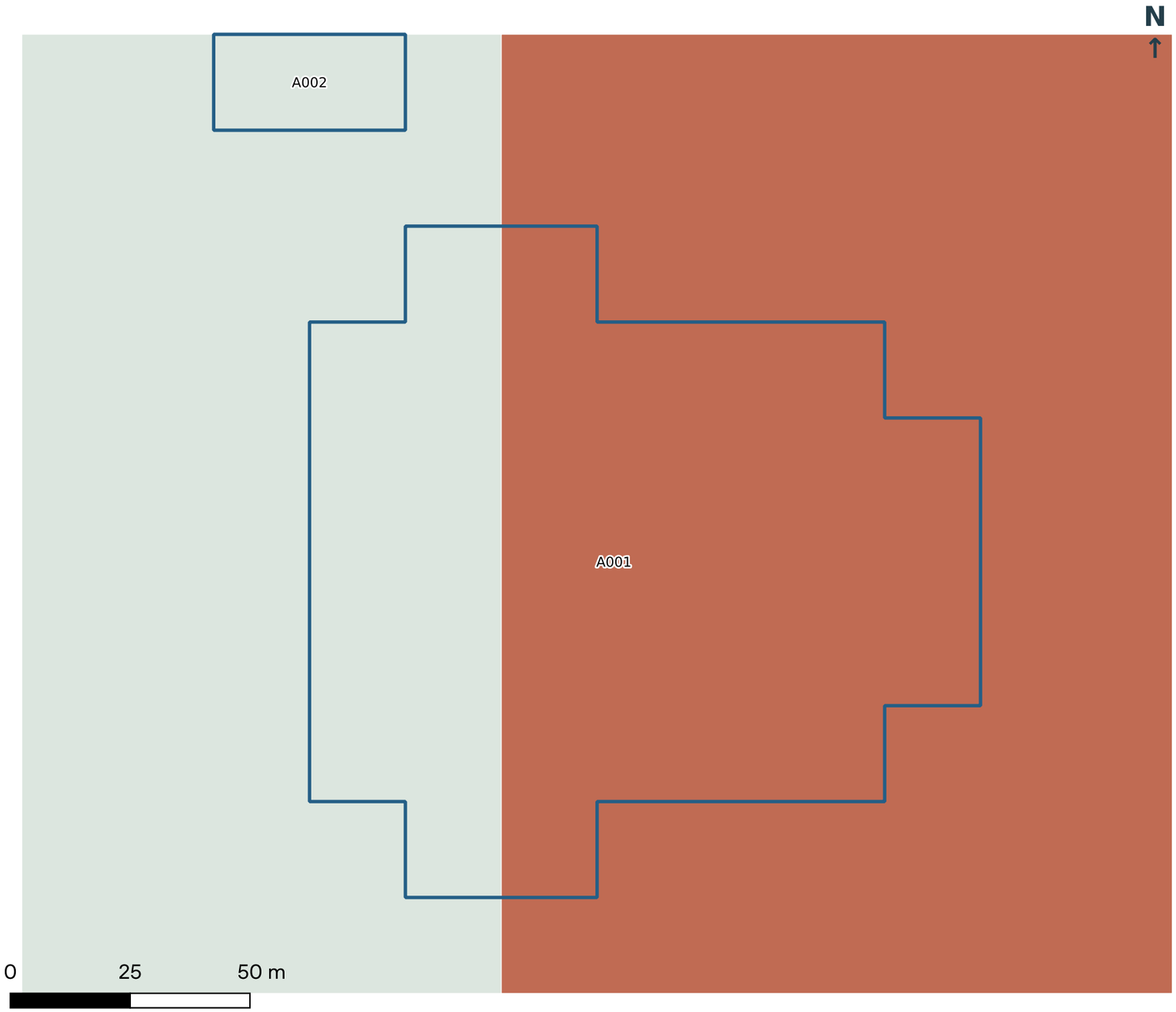
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: 20 × 20 m; source grid retained
Vertical datum: UNCONFIRMED
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_PLANNING_SRI5_V05
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

05 · Flood and heat planning context

Synthetic test domain · not actual Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

- Documented heat-priority context
- Other documented heat class

READ THIS MAP

Explicit source heat classes.
Blue outlines: flood action areas.
Priority overlap is recorded for
each flood action area.

This is baseline heat targeting.
No temperature reduction is claimed.
Thesis §§4.2, 5.7.4.

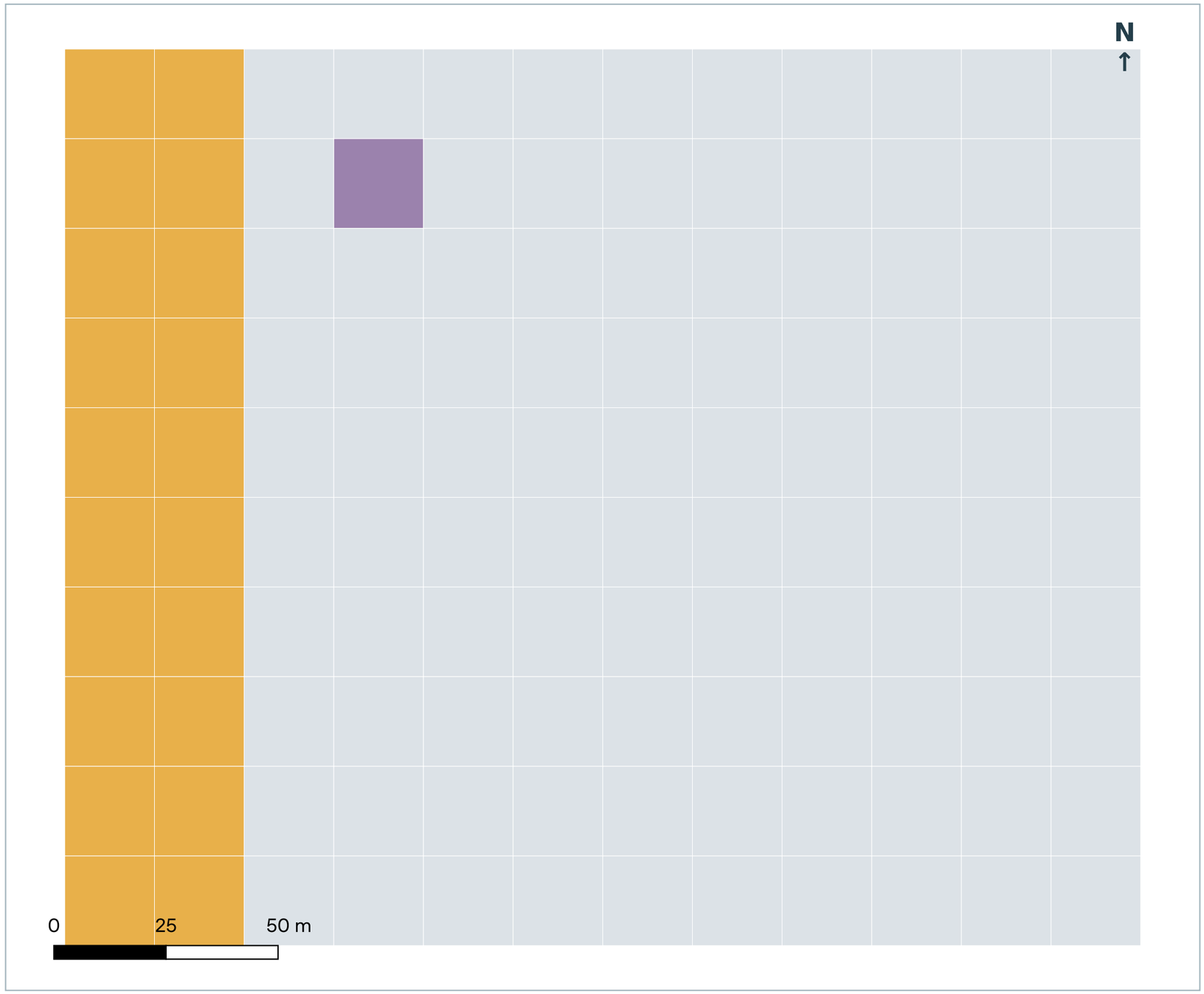
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: 20 × 20 m; source grid retained
Vertical datum: UNCONFIRMED
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_PLANNING_SRI5_V05
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

Roof retention / greening

Synthetic test domain · not actual Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

- Investigate · conditional planning concept
- Surface class missing / mixed
- Recorded surface does not match

READ THIS MAP

Surface match identifies an investigation opportunity only.
All retained concepts are conditional.

Grey: incompatible recorded surface.
Purple: surface evidence missing.
Known infiltration FAIL is retained in the pathway field.
Thesis Table 4.2; §§5.5.1–5.5.4.

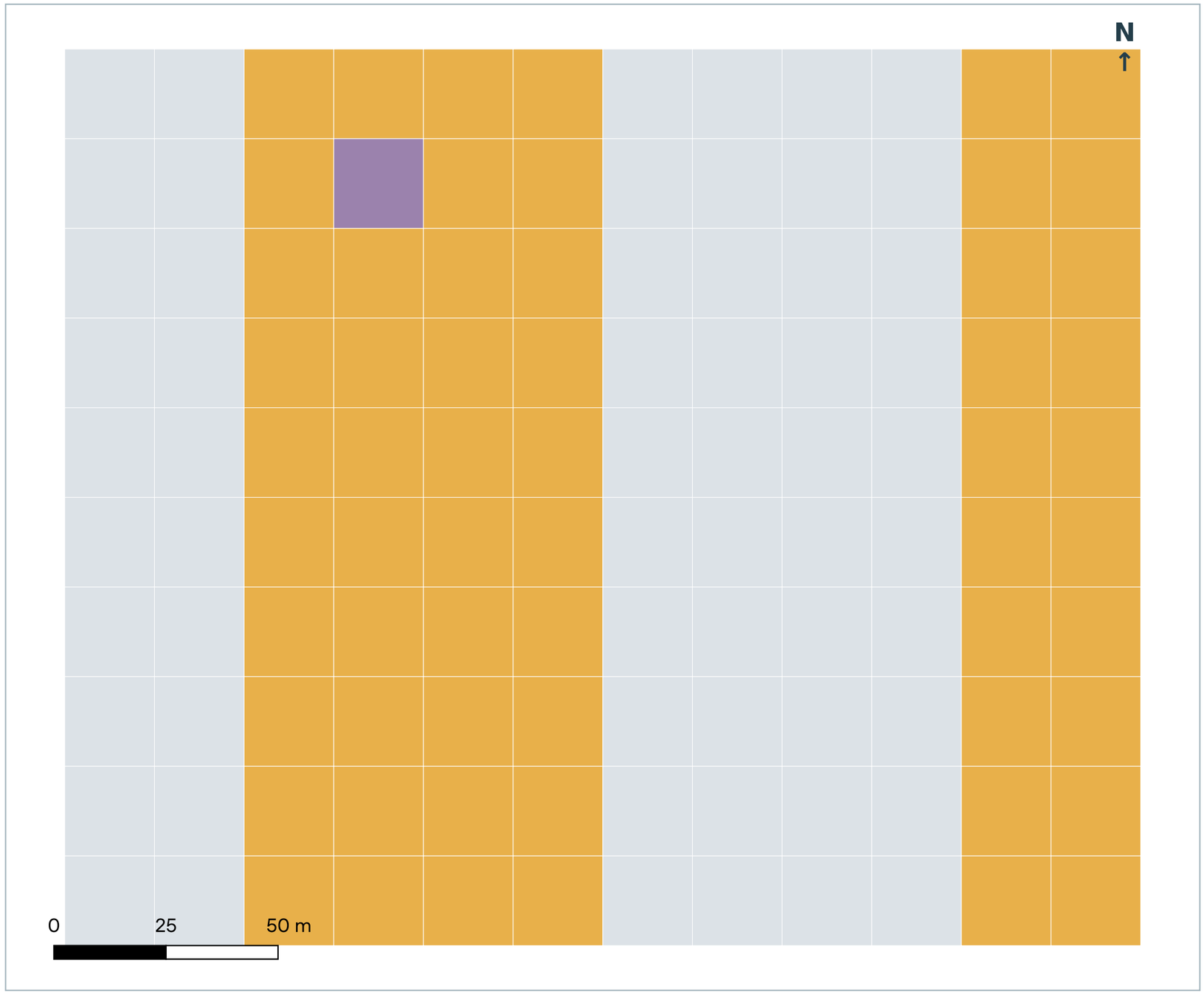
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: 20 × 20 m; source grid retained
Vertical datum: UNCONFIRMED
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_PLANNING_SRI5_V05
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

Rain gardens / planted detention

Synthetic test domain · not actual Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

- Investigate · conditional planning concept
- Surface class missing / mixed
- Recorded surface does not match

READ THIS MAP

Surface match identifies an investigation opportunity only.
All retained concepts are conditional.

Grey: incompatible recorded surface.
Purple: surface evidence missing.
Known infiltration FAIL is retained in the pathway field.
Thesis Table 4.2; §§5.5.1–5.5.4.

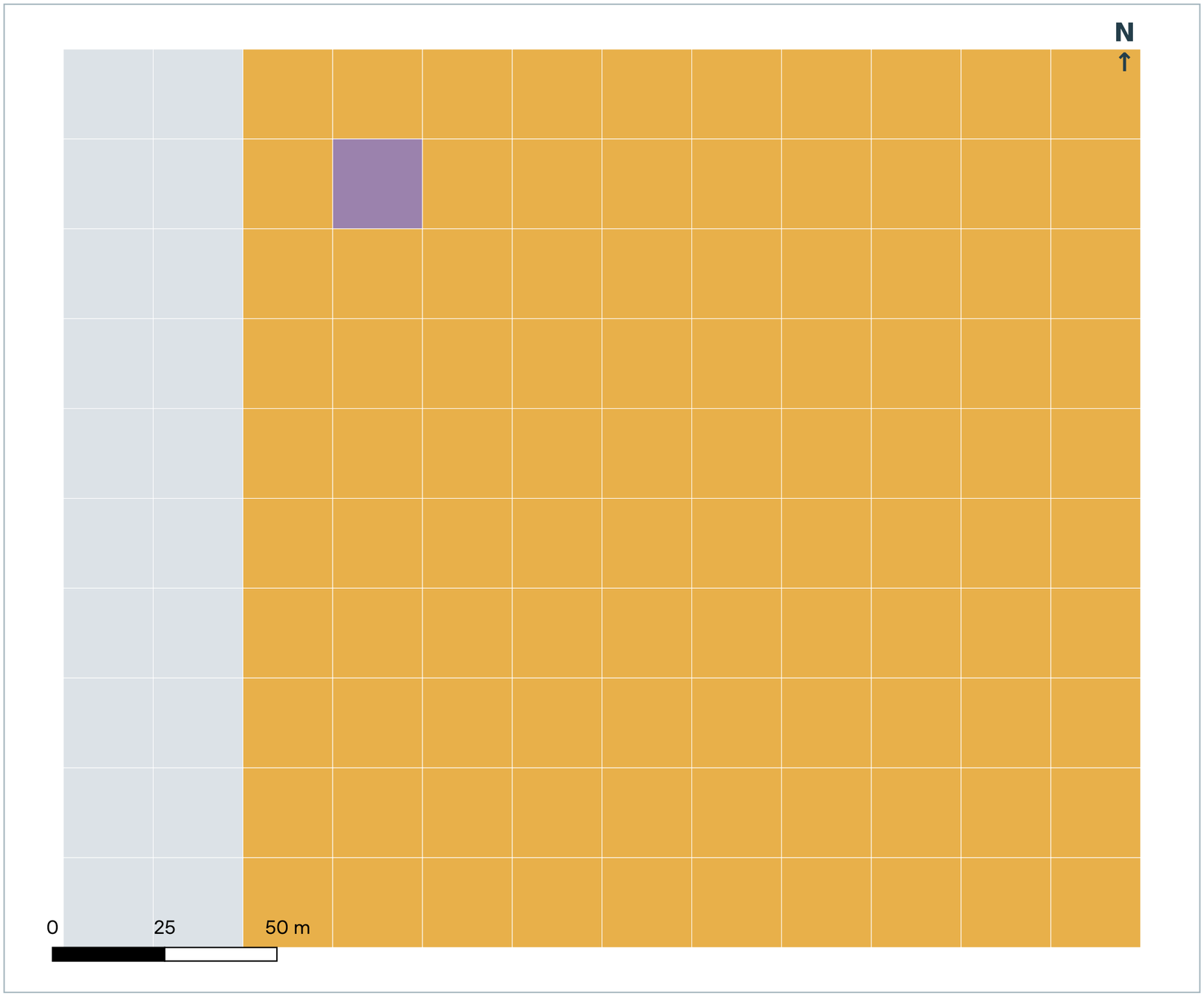
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: 20 × 20 m; source grid retained
Vertical datum: UNCONFIRMED
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_PLANNING_SRI5_V05
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

Water-sensitive trees / street retrofit

Synthetic test domain · not actual Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

- Investigate · conditional planning concept
- Surface class missing / mixed
- Recorded surface does not match

READ THIS MAP

Surface match identifies an investigation opportunity only.
All retained concepts are conditional.

Grey: incompatible recorded surface.
Purple: surface evidence missing.
Known infiltration FAIL is retained in the pathway field.
Thesis Table 4.2; §§5.5.1–5.5.4.

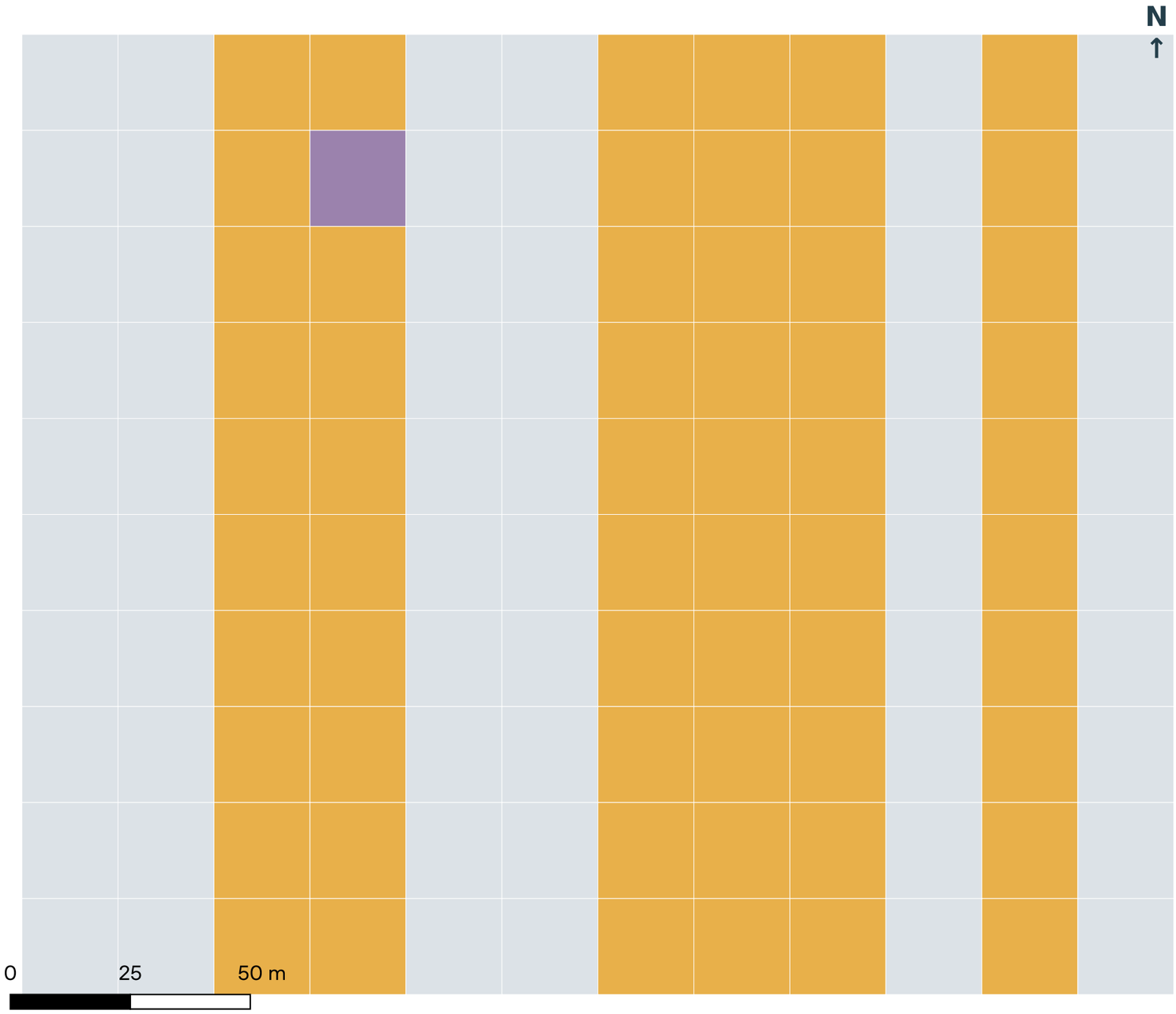
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: 20 × 20 m; source grid retained
Vertical datum: UNCONFIRMED
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_PLANNING_SRI5_V05
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

Multifunctional detention space

Synthetic test domain · not actual Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

- Investigate · conditional planning concept
- Surface class missing / mixed
- Recorded surface does not match

READ THIS MAP

Surface match identifies an
investigation opportunity only.
All retained concepts are conditional.

Grey: incompatible recorded surface.
Purple: surface evidence missing.
Known infiltration FAIL is retained
in the pathway field.
Thesis Table 4.2; §§5.5.1–5.5.4.

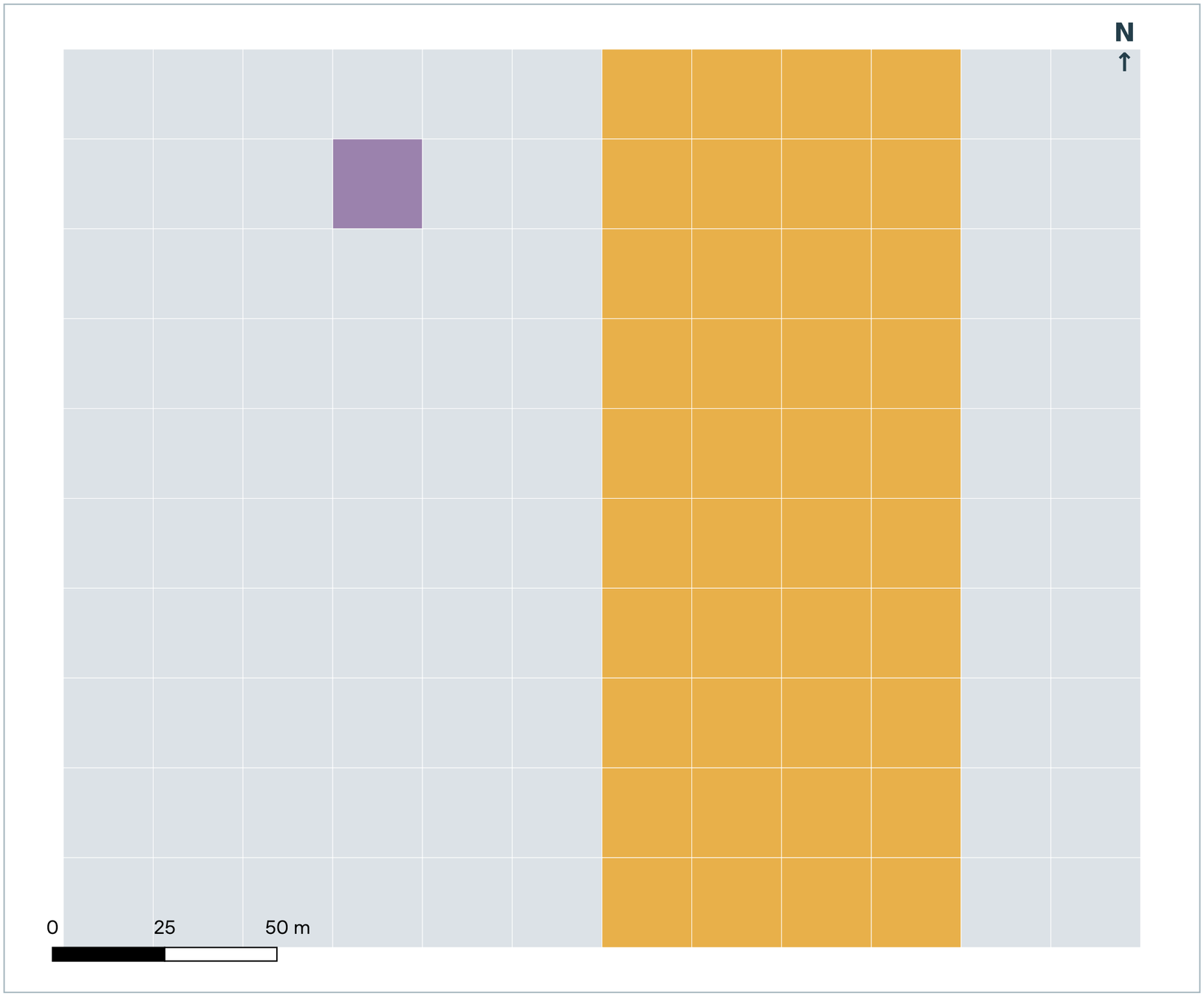
Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: 20 × 20 m; source grid retained
Vertical datum: UNCONFIRMED
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_PLANNING_SRI5_V05
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

Permeable / underdrained surfaces

Synthetic test domain · not actual Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

- Investigate · conditional planning concept
- Surface class missing / mixed
- Recorded surface does not match

READ THIS MAP

Surface match identifies an investigation opportunity only.
All retained concepts are conditional.

Grey: incompatible recorded surface.
Purple: surface evidence missing.
Known infiltration FAIL is retained in the pathway field.
Thesis Table 4.2; §§5.5.1–5.5.4.

Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: 20 × 20 m; source grid retained
Vertical datum: UNCONFIRMED
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_PLANNING_SRI5_V05
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Synthetic inputs; all evidence and authority decisions are invented for software testing.